

Transformers

Overview

- outline the insights and applications of Transformer models
- implement pre-training and fine-tuning

Async

Looking Back at RNNs

- form of f_{RNN}

$$h_t = \sigma[W_1 h_{t-1} + W_2 x_t + b_1]$$

- RNN is defined by ' W_1, W_2 , and b_1 '

- form of f_{pred}

$$o_t = \text{softmax}[W_3 h_t]$$

- encoder-decoder RNNs
 - two RNNs paired together

Design Hyper-Parameters

- L : number of layers
- H : number of attention heads
- d_{qk} : dimensionality of query and key vectors (" d_{qk} is d' ")
- d_v : dimensionality of value vectors (usually the same as d_{qk} " d ")
- d_{model} : dimensionality along residual path
- d_{ff} : dimensionality inside FNN
- Usually:

$$d_{model} = H d_v$$

$$d_{ff} = 4 d_{model}$$

Encoder Only Transformers	• process input sequences without generating outputs • excel in tasks such as text classification and sentiment analysis
Bidirectional Encoder Representations From Transformers (BERT)	• earlier language models (like the original GPT) read text sequentially—either left-to-right or right-to-left • this created limitations when the meaning of a word depended on the words following it • BERT solves this by analyzing words from both sides using a self-attention mechanism, providing a deeper, human-like understanding of context • pre-training tasks ○ masked language modeling (MLM) ▪ model randomly masks a portion of the input words and is trained to predict the missing words ○ next sentence prediction (NSP) ▪ predicts whether two sentences appear sequentially in a document, helping the model grasp long-range dependencies and relationships between texts • architecture ○ unlike models that use both encoders and decoders for text generation, BERT uses an encoder-only transformer stack • fine tuning ○ task selection

	▪ choose whether your downstream task requires sequence-level outputs (e.g., classifying a whole email as spam) or token-level outputs ○ architecture adjustment ▪ remove the pre-training heads and attach a minimal, task-specific output layer (such as a multi-layer perceptron) directly on top of BERT's encoder outputs ○ training ▪ feed your labeled dataset into the model ▪ during this phase, you update the newly added output layer and adjust (fine-tune) the pre-trained BERT weights to minimize prediction error ○ learning rate ▪ fine-tuning generally requires much smaller learning rates (e.g., 2×10^{-5} to 5×10^{-5}) than pre-training to prevent "catastrophic forgetting" of the previously learned language knowledge ○ batch size ▪ using slightly larger batch sizes (like 16 or 32) can stabilize gradient updates on smaller datasets
Decoder Only Transformers	• focus on generating output sequences • most suitable for language modelling tasks such as text generation and completion • GPT is a decoder-only transformer model • analysis is of generated words only, differing from the bidirectionality of BERT

	uses transformer stack for language generation																									
How to Train a Decoder-Only Transformer Stack	- ignore attention scores for future tokens in self-attention - set upper-triangular portion of the QK^T matrix to $-\infty$ <table><tr><td>$q_1 \cdot k_1$</td><td>$-\infty$</td><td>$-\infty$</td><td>$-\infty$</td><td>$-\infty$</td></tr><tr><td>$q_2 \cdot k_1$</td><td>$q_2 \cdot k_2$</td><td>$-\infty$</td><td>$-\infty$</td><td>$-\infty$</td></tr><tr><td>$q_3 \cdot k_1$</td><td>$q_3 \cdot k_2$</td><td>$q_3 \cdot k_3$</td><td>$-\infty$</td><td>$-\infty$</td></tr><tr><td>$q_4 \cdot k_1$</td><td>$q_4 \cdot k_2$</td><td>$q_4 \cdot k_3$</td><td>$q_4 \cdot k_4$</td><td>$-\infty$</td></tr><tr><td>$q_5 \cdot k_1$</td><td>$q_5 \cdot k_2$</td><td>$q_5 \cdot k_3$</td><td>$q_5 \cdot k_4$</td><td>$q_5 \cdot k_5$</td></tr></table> this is called “causal masking” - apart from causal masking, no difference from the encoder-only architecture IMPORTANT: unlike RNN LMs, this training can be completely parallelized	$q_1 \cdot k_1$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$q_2 \cdot k_1$	$q_2 \cdot k_2$	$-\infty$	$-\infty$	$-\infty$	$q_3 \cdot k_1$	$q_3 \cdot k_2$	$q_3 \cdot k_3$	$-\infty$	$-\infty$	$q_4 \cdot k_1$	$q_4 \cdot k_2$	$q_4 \cdot k_3$	$q_4 \cdot k_4$	$-\infty$	$q_5 \cdot k_1$	$q_5 \cdot k_2$	$q_5 \cdot k_3$	$q_5 \cdot k_4$	$q_5 \cdot k_5$
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Advantages of Transformer-Based Generation	- parallelized training can enable larger models and training sets - training is stable compared to RNNs - dur to layer normalization and residual connections - no vanishing or exploding gradients, “path length” is always = 1																									
GPT Series of LMs	GPT = Generative Pretrained Transformers <table><tr><th>Model</th><th>Parameters</th><th>Decoder Layers</th><th>Context Window Size</th><th>Hidden Layer Size</th></tr><tr><td>GPT-1</td><td>117 M</td><td>12</td><td>512</td><td>768</td></tr><tr><td>GPT-2</td><td>1.5 B</td><td>48</td><td>1024</td><td>1600</td></tr><tr><td>GPT-3</td><td>175 B</td><td>96</td><td>2048</td><td>12288</td></tr><tr><td>GPT-4</td><td>Unknown</td><td>Unknown</td><td>Unknown</td><td>Unknown</td></tr></table>	Model	Parameters	Decoder Layers	Context Window Size	Hidden Layer Size	GPT-1	117 M	12	512	768	GPT-2	1.5 B	48	1024	1600	GPT-3	175 B	96	2048	12288	GPT-4	Unknown	Unknown	Unknown	Unknown
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**Encoder-
Decoder
Transformers**

- this was the original transformer architecture
 - machine translation / summarization
- example
 - “even when **he** was a young boy, Rich Sanchez...”
 - identifies the pronoun precedes the subject that it references
- T5 (Text-to-Text Transfer Transformer)
 - unified approach where all NLP tasks are reformulated as text-to-text tasks
 - example
 - “classify sentiment: the movie was horrendous...”
- BART (Bidirectional and Auto-Regressive Transform)
 - combines bidirectional and auto-regressive training objective
 - good with both generation and sequence-to-sequence tasks